AGROSPEAR CROPCARE CO., LTD.

HERBICIDE MANUFACTURING QUALITY CONTROL SYSTEM & DOSSIER

除草剂制剂生产全流程质量控制体系与佐证报告规程

Document Code: SOP-QC-HERB-2026
Effective Date: August 2026
Version: Rev 4.0 (Global Standard)
Compliance: FAO/WHO Specs & CIPAC Methods
Product Lines: SL, SC, OD, EC, WDG Herbicides

MANDATORY HERBICIDE SAFETY PROTOCOL: Herbicides must be manufactured in strictly dedicated, segregated workshops with negative pressure HVAC and CIP (Clean-in-Place) validation. Cross-contamination limit for sulfonylureas/auxin mimics in non-target agrochemical lines is < 10 ppb (LC-MS/MS verified).

除草剂生产核心安全强制规程： 除草剂生产必须在物理完全隔离的专用车间内进行，配备负压排风与洁净 CIP 自动清洗系统。针对超高效除草剂（磺酰脲类、激素类等），对非目标作物制剂生产线的交叉污染极限值严格控制在 **< 10 ppb**（经 LC-MS/MS 超痕量质谱验证）。

1. END-TO-END HERBICIDE QA/QC WORKFLOW / 除草剂生产品控全生命周期流程

QC Stage / 阶段	Key Control Parameters & Protocols / 关键管控项目与规范	Methodology / 检验方法与仪器	Release Criteria / 准则
IQC (Incoming QC) **原辅料进厂检验**	**• Technical Material (TC/TK): AI purity, toxic relevant impurities (e.g., free glyphosate/AMPA, hydrazine), isomer ratios (e.g., S-Metolachlor enantiomer ee ≥ 88%), moisture.** • 原药(TC/TK): 有效成分纯度、限制性有害杂质（如游离甲醛/AMPA、肼类限度）、光学异构体比例（如精异丙甲草胺 S-体光学纯度 ee ≥ 88%）与水分。 **• Solvents & Adjuvants: Methyl oleate (biodiesel carrier), tallow amine ethoxylates, silwet surfactants, oil absorption value for solid carriers.** • 油相载体与助剂: 甲基化植物油载体酸价与水分、牛脂胺聚醚/有机硅表面张力、载体白炭黑吸油值。 **• Dedicated Packaging: Co-extruded fluorinated HDPE bottles, anti-glug necks, torque leak testing.** • 专用包材: 高阻隔氟化 HDPE 瓶、防浪涌瓶口、耐摔抗跌落及铝箔感应封口密封性。	• Chiral HPLC / GC-FID • Karl Fischer Titrator • Surface Tensiometer • Vacuum Sealing Tester	100% Passed 不合格严禁入库
IPQC (In-Process) **制程过程巡检**	**• Soluble Liquid (SL): Neutralization reaction pH endpoint (e.g., Glufosinate-ammonium / Glyphosate IPA), salt conversion rate > 99.5%, clarity.** • 水剂(SL): 中和成盐反应 pH 终点判定（草铵膦铵盐/草甘膦异丙胺盐）、转化率 > 99.5% 与水溶澄清度。 **• Oil Dispersion (OD): Bead milling chamber temp < 32°C (prevent sulfonylurea degradation), moisture control < 0.3%, 3D structuring network for non-settling.** • 可分散油悬浮剂(OD): 研磨控温 < 32°C（防磺酰脲类热解）、严控体系微量水 < 0.3%、构建立体抗沉降网络。 **• Emulsifiable Conc. (EC): Esterification stability, low-temp crystallization monitoring at -5°C.** • 乳油(EC): 酯类除草剂相容性、-5°C 极低温耐寒结晶巡查。 **• Filling Line: Volumetric accuracy (±0.5%), torque verification, 2D QR serialization.** • 灌装线: 装量称重监控、瓶盖扭矩测试、高频封口与一物一码追溯喷码。	• Online pH & Viscometer • Laser Diffraction (D50/D90) • Coulometric KF Titrator • Density Meter	CCP In-Spec 偏离立即停线
FQC / OQC **成品终检与放行**	**• Multicomponent Herbicide Assay: Baseline chromatographic separation (e.g., Glufosinate + Oxyfluorfen / Mesotrione + Atrazine).** • 多元复配除草剂有效成分定量分析: HPLC 离子对/反相色谱完全分离，符合 FAO 允差。 **• Physicochemical: Dispersion stability in standard water (CIPAC MT 180), Persistent foam (CIPAC MT 47.2 ≤ 25mL), Pourability residue ≤ 2.5%.** • 关键理化性质: 油悬浮分散稳定性、起泡量 ≤ 25mL、倾倒洗涤残留 ≤ 0.3%、低温冷储无分层析油。 **• Accelerated Thermal Stability: 54±2°C for 14 days (Sulfonylureas at 45°C) → Active degradation ≤ 3.0%.** • 热储加速破坏试验: 54°C 热储 14 天（磺酰脲类执行 45°C 特殊热储），有效成分降解率 ≤ 3.0%。	• CIPAC MT 180 / MT 184 • CIPAC MT 46.3 / MT 39.3 • LC-MS/MS Trace Cleanness	COA Released QA 总监审批放行
Traceability & Cleaning **清场验证与追溯**	**• Mandatory CIP Swab Validation: HPLC/MS residue test on tank walls before switching herbicide batches (< 0.1 ppm).** • 车间清场与清洗验证: 换批次必须实施 CIP 碱洗与有机溶剂冲洗，擦拭取样化验残留（< 0.1 ppm）。 **• Retained samples archived at 15–25°C for shelf-life + 1 year (min. 3 years) with batch manufacturing records (BMR).** • 阴凉留样库留存至保质期后 1 年，配套完整批生产指令单（BMR）与实验室图谱归档。	• Swab Sampling LC-MS • Cloud Traceability ERP	Archived 建立双向追溯台账

2. FORMULATION-SPECIFIC QC STANDARDS (FAO/CIPAC COMPLIANT) / 主要除草剂剂型质控指标

Formulation / 剂型	Representative Products / 代表品种	Critical Quality Attributes (CQA) / 核心理化质控指标	Key Test Method / 对应方法
SL (Soluble Liquid) **可溶性水剂**	• Glufosinate-ammonium 200g/L SL • Glyphosate 41% / 480g/L IPA SL • 2,4-D Dimethylamine 720g/L SL	• Active Content: Nominal ± 5.0% • pH Range: 5.0 – 8.0 (1% aqueous solution) • Water Solubility & Clarity: 100% transparent • Persistent Foam: ≤ 20 mL after 1 min	• HPLC / Titration • CIPAC MT 75.3 • CIPAC MT 47.2
OD (Oil Dispersion) **可分散油悬浮剂**	• Nicosulfuron 40g/L OD • Mesotrione 5% + Atrazine 20% OD • Penoxsulam 25g/L OD	• Moisture Content: ≤ 0.30% (w/w) • Particle Size: D50 ≤ 1.8µm, D90 ≤ 4.0µm • Dispersion Stability: Phase separation ≤ 2.0 mL • Thermal Stability (45°C, 14d): AI degradation ≤ 3.5%	• CIPAC MT 180 • Karl Fischer MT 30.5 • Laser Diffraction
EC (Emulsifiable Conc.) **乳油**	• Quizalofop-p-ethyl 10% EC • S-Metolachlor 960g/L EC • Clethodim 240g/L EC	• Optical Purity (ee value): ≥ 88.0% (for S-Metolachlor) • Moisture Content: ≤ 0.35% • Emulsion Stability: 0.5h/2h/24h no separation • Low Temp. Stability: 0±2°C for 7 days (clear)	• Chiral HPLC • CIPAC MT 36.3 • CIPAC MT 39.3
SC / WDG (Suspension/ Granule) **悬浮剂 / 水分散粒剂**	• Atrazine 500g/L SC • Bispyribac-sodium 80% WDG • Pyrazosulfuron-ethyl 10% WP/WDG	• Suspensibility: ≥ 92.0% (SC) / ≥ 88.0% (WDG) • Disintegration Time: ≤ 60 seconds (WDG) • Wet Sieve Residue (75µm): ≥ 99.0% through • Hardness & Friability: Attrition resistance ≥ 95%	• CIPAC MT 184 • CIPAC MT 174 • CIPAC MT 178.2

3. SUPPORTING EVIDENCE 1: OFFICIAL COA TEMPLATE / 佐证文件一：成品出厂检验报告单 (COA)

AGROSPEAR CROPCARE CO., LTD.

CERTIFICATE OF ANALYSIS / 产品检验分析报告单 (HERBICIDE)

Product Name / 品名: Glufosinate-ammonium 200g/L SL (精草铵膦/草铵膦 200g/L 水剂)

Batch No. / 生产批号: AS-HERB-20260826-08

Invoice / Order No. / 订单号: AGR-EXP-2026-928

Batch Quantity / 批量: 15,000 Liters (1L x 12/Carton)

Manufacture Date / 生产日期: 2026-08-26

Expiry Date / 有效期至: 2028-08-25

Sampling Date / 取样日期: 2026-08-26

Inspection Date / 判定日期: 2026-08-27

Test Parameter / 检验项目	Specification Limit / 标准指标	Test Result / 实测结果	Standard Method / 检验方法	Verdict / 判定
Appearance / 外观性状	Clear blue or transparent homogeneous liquid (澄清均质液体)	Conforms / 符合要求	Visual Check	Pass
Active Ingredient: Glufosinate-ammonium	200.0 ± 10.0 g/L (190.0 – 210.0 g/L)	**201.5 g/L**	HPLC-UV (Agilent 1260)	Pass
pH Value (1% aqueous solution)	5.0 – 8.0	6.5	CIPAC MT 75.3	Pass
Density (20°C) / 密度	1.080 – 1.120 g/cm³	1.102 g/cm³	CIPAC MT 3.2.1	Pass
Water Solubility & Solution Stability	Completely clear, no sediment / oil slick	Clear / 完全澄明	CIPAC MT 41	Pass
Persistent Foaming / 持久起泡性	≤ 20 mL after 1 min	8 mL	CIPAC MT 47.2	Pass
Low Temp. Stability (0±2°C, 7 days) / 冷储	No solid precipitation or phase separation (无析出)	Conforms / 澄清无沉淀	CIPAC MT 39.3	Pass
Heat Stability (54±2°C, 14 days) / 热储	AI Degradation ≤ 3.0%, pH drift ≤ 0.5	Degradation: 0.7%, pH: 6.4	CIPAC MT 46.3	Pass
Cross-Contamination Clean Check / 残留排查	Sulfonylurea / Triazine active residues < 0.05 ppm	Not Detected / 未检出	LC-MS/MS	Pass

CONCLUSION / 检验结论: PASS - Fully complies with FAO Specifications & Enterprise Quality Standard. Approved for release. (合格 - 各项理化指标均符合 FAO 规格及企业内控标准，准予放行出厂。)

Analyst / 检测员:
Zhang Wei (张伟)
Date: 2026-08-27

Reviewer / 复核员:
Li Ming (李明)
Date: 2026-08-27

QA Director / QA总监:
Dr. Alex Chen (陈博士) *[Approved]*
Date: 2026-08-27

4. SUPPORTING EVIDENCE 2: BATCH RELEASE & OOS/CIP PROTOCOL / 佐证文件二：批放行审批与超标/清场规程

Release Gate / 质量关卡	Verification Requirements / 审核项目	Status / 状态	Verified By / 审核人
1. Raw Material Gate (IQC)	Active ingredient assay & isomer purity (ee%), carrier moisture < 0.3%, fluorinated bottle barrier check	Verified Pass	IQC Lead
2. Process Parameters (IPQC)	SL neutralization pH endpoint, OD milling temp < 32°C, CIP cleaning swab residue LC-MS check passed	Verified Pass	IPQC Officer
3. Laboratory FQC (COA)	Active chromatography quantitative assay, solution clarity, thermal/cold stability, zero cross-contamination	Verified Pass	Lab Supervisor
4. Packaging & Traceability	2D QR code serialization, high-frequency inductive seal 100% leak check, carton drop integrity	Verified Pass	Packaging QA

HERBICIDE OOS & CROSS-CONTAMINATION INVESTIGATION: If any batch fails active content or shows cross-contamination signals, the production line is immediately locked with physical Red Tags. Re-homogenization or rework must undergo a 3-tier safety audit and requires joint sign-off by the Factory Director and Quality Assurance Director.

除草剂超标（OOS）与交叉污染调查处置规程： 凡任一批次出现有效成分偏差或交叉污染可疑信号，系统自动锁定出库闸口并张贴物理红卡。任何重调、返工或报废方案必须执行严格的三级安全审计，并经工厂厂长与质量总监双重书面签字批准后方可实施。